

- Types of display systems $\Rightarrow$

- $\rightarrow$ Raster Display system

- $\rightarrow$ Vector Display system

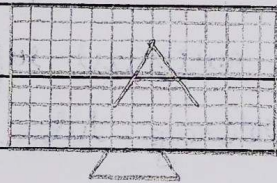
- $\rightarrow$ Pixel Plotter.

- $\rightarrow$ Point Plotter.

- Raster Display system $\Rightarrow$

- $\rightarrow$ Uses a rectangular grid of pixel image formed by

- E.g. $\Rightarrow$ LCD, LED monitors.



- Vector Display system $\Rightarrow$

- $\rightarrow$ Draw lines directly using an e-beam between two points.

- E.g. $\Rightarrow$ early CAD software.

- Pixel Plotter $\Rightarrow$

- $\rightarrow$ Each pixel is individually addressed and illuminated.

- E.g. $\Rightarrow$ used in raster graphics where precision is needed.

- **Point Plotter** $\Rightarrow$
 - $\rightarrow$ Plot a single point rather than complete line or shape.
 - $\rightarrow$ e.g. $\Rightarrow$ used in scientific plotting system.
- **Display Memory Technology** $\Rightarrow$
 - 1) **Continual Refresh Display:**
 - $\rightarrow$ Requires constant refreshing of the image from the memory.
 - 2) **Storage Display:**
 - $\rightarrow$ Image is retained without refreshing.
 - e.g. $\Rightarrow$ Direct view storage tube.
 - 3) **Digital Frame Buffer:**
 - $\rightarrow$ Memory buffer storing pixel data for rendering.
 - 4) **Hardware specific Display: Very High Resolⁿ Display:**
 - $\rightarrow$ Display devices with very high ~~subpixel~~ pixel density e.g.: 8K, 4K display.
 - 5) **Plasma Panel Display** $\Rightarrow$
 - $\rightarrow$ Ionized gas emits light between two gas panels (Xe, Ne) used in flat screen TV industrial display.

- **Unit 1 $\Rightarrow$ ~~line~~ and point pl Display devices:**
 - $\rightarrow$ line and point plotting system, Raster vector pixel and point plotter, continual refresh and storage display, digital frame buffer, plasma panel display, very high resolution devices, high speed drawing, display processor, character generator colour display techniques.
- $\rightarrow$ used in medical imaging, simulation.
- **High speed Drawing** $\Rightarrow$
 - $\rightarrow$ It refers to the ~~graphic~~ ^{rapid} generation of images (specially line, shapes and objects) on display device, typically required in real time or real-time applications.
 - $\rightarrow$ Application such as simulation, animation, CAD software, gaming.
 - $\rightarrow$ It emphasise both speed and responsiveness of graphics.
- **Digital Frame Buffer** $\Rightarrow$
 - $\rightarrow$ It is a dedicated block of memory that stores pixel data for image display.
 - $\rightarrow$ Each memory ~~stores~~ ^{buffer} corresponds to a pixel on a screen.
 - $\rightarrow$ Holding information about its colour, intensity and other attributes.

→ Frame buffer enables smooth rendering of images, animations and videos in dynamic and static display.

• what is static and dynamic display.

• Types of digital frame buffer:

1) Single frame buffer → stores only one frame of image data.

e.g. → images.

2) Double frame buffer → uses two frame buffers one for current and other for next preparing the next frame.

3) Triple buffering → adds a third buffer to further reduce latency and tearing in high performance graphics rendering.

4) Depth Buffer (Z buffer) → stores the depth infoⁿ for each pixel in 3D rendering.

e.g. Gen AI images.

• working of Digital Frame Buffer →

→ Memory allocation: memory is reserved for frame buffer with its size determined by the resolution and color depth.

e.g. → 24 bit.

→ Image data storage → pixels information like color, brightness, transparency is ^{stored} in buffer for refreshing.

→ Rendering

• Display Processor →

→ It is a specialized hardware component that manage the rendering and output of images, videos, graphical content to a display device. It handle tasks like image scaling, color conversion, frame buffering to off load work from CPU or GPU.

→ Ensuring smooth and efficient graphical performance.

• Types:

1) General Purpose GPUs.

2) Digital Signal Processor

3) Image Signal Processor.

4) video Processing unit.

5) Embedded display processor.

1) General purpose GPU: perform a wide range of tasks including 2D and 3D image

e.g. → NVIDIA GeForce

2) Digital signal ^{processor} specialized for real time image and video processing.

→ Designed for processing raw image data from camera and sensors.

→ Handles video decoding encoding and streaming. e.g. → ~~medical~~ medical.

→ Embedded display processor: found in devices like TVs, projectors and phones for scaling and color enhancement.

working: input data received from CPU GPU or external devices such as camera, perform tasks like image scaling, noise reduction.

• Frame Rate Adjustment: memory management use frame buffers to temporarily store before outputting them to the display.

→ sends the signal to the display device such as monitor, VR headset.

• Optimisation:

→ Reduce load on the main CPU or

GPU by taking over specific tasks improving system efficiency.

• Advantages →

→ Improve performance.

→ Enhance quality.

→ ~~Energy efficient~~

→ Real time processing

→ versatility.

• Disadvantages →

→ High cost.

→ High power consumption

→ compatibility issue.

• Character Generator →

→ It is a software application that produce static or animated text for used in 2D or 3D videos.

→ It can be used to create anything from simple lower third text to full blown 3D animation.

• working of character generator:

→ It is a software that creates graphical images and animation for use in video production. The images are usually created

from scratch by a team of artist or they maybe taken from pre-existing database of images.

→ The animations are created by an animator who designs the movement of the character and objects in the scene.

→ The character generation is used to generate the images and animation that are then combined with live action footage or other graphics to create a final production.

→ It is used in TV or broadcast or film or video games, ~~or air~~

→ Animated texts such as credit rolls, news heading.

• Advantages:

→ versatility

→ scalability.

→ Real time capability.

→ Memory efficiency.

→ Customize

• Disadvantage:

→ complexity

→ Resource usage